ChatGPT

Non-Parametric In-Context Depth Estimation

Architectural Precedents

Several recent works incorporate support images or retrieved data to improve monocular depth. For example, RAD (Retrieval-Augmented Depth) retrieves a context RGB–depth pair and fuses it with the query via a dual-stream network and a *matched cross-attention* mechanism [1]. In RAD, spatial correspondences between the query and support guide cross-attention so that depth information is transferred only at matching points [1]. Likewise, meta-learning or retrieval-based methods (e.g. Hummingbird) use nearest-neighbor retrieved examples for depth without weight updates [2] [3]. Many of these systems operate non-parametrically: they encode images with a frozen transformer and perform retrieval or attention at inference time, effectively using the support depth as an external reference. By contrast, traditional depth models embed geometry solely in parameters; these new designs explicitly inject depth context via cross-image attention. (Note that multi-view or stereo networks also relate multiple frames, but they require known baselines or pose.)

Comparison to Visual-Prompting Approaches

Our idea is also related to recent "visual prompting" or in-context architectures, but with key differences. Generalist models like Painter use image–image prompts to "inpaint" task outputs: for depth tasks, Painter achieved SOTA on NYUv2 by training a massive ViT on paired images and outputs [4]. Painter's prompts are task-indicating image pairs rather than ground-truth depth maps. Similarly, Bar et al. propose an MAE+VQGAN inpainting model that can be "prompted" by example input–output images [5]. These methods treat depth as just another image in an inpainting framework, relying on large pretraining and discrete token masking. By contrast, our approach uses *actual* support images with real depth ground truth as context, and a much smaller transformer that attends to both support and query. Other depth-specific prompt methods (e.g. PromptMono) use *learnable* prompt tokens and gated cross-attention to inject domain cues [6]. We instead feed real depth maps to calibrate scale. In summary, unlike general prompting models that unify many tasks with one big model, our design targets continuous depth output and explicitly uses a reference depth to guide prediction, which is novel to dense regression.

Functional Comparisons

- *Context windows of ground-truth frames:* To our knowledge no prior work has directly fed a sliding window of true depth frames into a small transformer at inference. Instead, methods like Hummingbird retrieve a single annotated image and transfer its semantics non-parametrically [2]. In contrast, we imagine a *few-shot analog* of video-depth: one could conceive feeding multiple reference images with LiDAR depths to bolster a low-capacity model, but this has not been explored in the literature.
- *ICL vs. Fine-tuning sample efficiency:* Recent studies of in-context learning for vision suggest retrieval-based inference can be highly data-efficient. For example, Balažević et al. show their Hummingbird encoder with nearest-neighbor decoding rivals specialized models with far fewer examples, often

outperforming end-to-end finetuning in the low-shot regime [7]. In that work, kNN retrieval on <6 % of VOC data beats full finetuning, and similarly on ADE20K with <1 % of data [7]. This indicates that conditioning on a small context can adapt a model with minimal data, whereas finetuning requires more samples. We found no depth-specific benchmarks explicitly comparing in-context conditioning to finetuning for domain shifts; most depth adaptation (e.g. CLIP-based or self-supervised domain transfer) assumes weight updates or extra training, not purely in-context inference.

Inductive Bias and Conditioning

Transformers in vision are mathematically quite general. Formal analyses treat attention as a graph layer: standard self-attention is *permutation-equivariant* (no bias) and encoder–decoder (cross-attention) layers are *block-permutation-equivariant*, meaning they make no built-in geometric assumptions unless positional encodings are used [8]. In practice, a transformer learns any needed bias from data. We found no theory showing that adding a support depth map steers it toward relative matching; rather, the model must learn to use the depth context. Providing a depth map in the prompt effectively acts as additional *conditioning input* (similar to a bias term that varies by context) but does not alter the transformer's weights. It is conceptually closer to a dynamic prompt than a true hypernetwork: the network's weights are fixed, and the support simply modulates activations via cross-attention. We found no benchmark or study labeling this as hypernetwork or dynamic-bias; it is mostly a practical architectural choice.

State-of-the-Art Reference-Based Models

A few recent systems exemplify "reference-based" depth inference (no weight updates). **Hummingbird** (Balažević et al.) is a self-supervised ViT pretrained with cross-image attention; using nearest-neighbor retrieval of annotated image features, it can do segmentation and depth "in-context" without fine-tuning, almost matching specialist models [2]. **RAD** (Baltaxe et al.) explicitly retrieves similar training images with depth and performs matched cross-attention, yielding large gains on rare classes [1]. **Painter** (Wang et al.) casts depth estimation as an image-inpainting task with an in-context prompt of input/output pairs; impressively, Painter achieved SOTA results on NYUv2 depth purely through its generalist masked modeling [4]. (Although Painter's model is huge and trained on many tasks, its inference is reference-based and weight-frozen for each task.) We are not aware of any published model that uses a dynamically supplied real depth map as in-context input in the way we propose; these nearest-neighbor and prompting systems are the closest analogues.

Key Differentiators

- **Arbitrary support:** Unlike stereo or video-based methods that use temporally adjacent frames, our support image can be any scene with similar geometry. Prior works mostly assume some fixed relationship between views (e.g. known pose or sequential frames), whereas here the support is "zero-shot" from anywhere.
- **Small-model utility:** Our goal is to let a lightweight model leverage external context. This shifts scale and memory burden from the weights into the prompt data. Large unified models (Painter, MAE-VQGAN, etc.) already excel at many tasks, but our method specifically aims to empower *low-capacity* networks to handle novel scenes by offloading priors to the support pair. In essence, the depth map in the context "teaches" the model on the fly, so even a small transformer can achieve strong accuracy without gigantic learned parameters.

Sources: Recent papers and preprints on depth estimation and visual prompting provide the basis for these comparisons [1] [4] [7] [6] [2], illustrating that while some works use retrieved or prompted images (Painter, Hummingbird, RAD), the specific idea of using an arbitrary support RGB–depth pair as a prompt in a small cross-attention model appears novel. The autonomous-driving context (e.g. few-beam LiDAR as in LiDARTouch [9]) shows related scale cues, but again with training-time integration rather than on-the-fly prompting. These references clarify the current landscape and highlight what is new in the proposed approach.

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[1] RAD: Retrieval-Augmented Monocular Metric Depth Estimation for Underrepresented Classes
https://arxiv.org/html/2602.09532v2

[2] [3] [7] [2306.01667] Towards In-context Scene Understanding
https://ar5iv.org/pdf/2306.01667

[4] Images Speak in Images: A Generalist Painter for In-Context Visual Learning
https://openaccess.thecvf.com/content/CVPR2023/papers/Wang_Images_Speak_in_Images_A_Generalist_Painter_for_In-Context_Visual_CVPR_2023_paper.pdf

[5] openreview.net
https://openreview.net/pdf?id=o4uFFg9_TpV

[6] [2501.13796] PromptMono: Cross Prompting Attention for Self-Supervised Monocular Depth Estimation in Challenging Environments
https://ar5iv.org/pdf/2501.13796

[8] Relational inductive biases on attention mechanismsOriginal version published in Spanish into the journal Rsearch in Computing Science https://www.rcs.cic.ipn.mx/
https://arxiv.org/html/2507.04117v1

[9] [2109.03569] LiDARTouch: Monocular metric depth estimation with a few-beam LiDAR
https://arxiv.org/abs/2109.03569